

Mapping Child Well-Being and Public Spending in Europe: a Non-Compensatory analysis for policy insights

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Abstract. This study examines child well-being across sixteen European countries in 2015 and 2018, using the OECD Child Well-being dataset. We apply a non-compensatory partial-order framework to preserve the ordinal structure of the indicators and avoid arbitrary trade-offs across dimensions. The analysis combines a two-step Mutual Ranking Probability cascade, dominance-based rankings, Minimum Dominating Set extraction, and separation-based visualisation. The results show that Sweden remains the only stable leader across both waves, while Estonia exits the frontier due to a marked decline in social well-being despite gains in some underlying educational indicators. More broadly, the study reveals that European child well-being is structured around multiple incomparable profiles rather than a single linear ranking, offering a clearer policy-oriented view than composite indices.

KEYWORDS: Partially Ordered Sets (POSets), Child Well-Being, Mutual Ranking Probability, Bubley-Dyer MCMC Sampling, Multidimensional Scaling.

1. DATASET

OECD Child Well-being (CWB) Dataset.

Source: Extracted via SDMX/CSV API protocol from the OECD Child Well-being Explorer (April 2026).

Link: OECD Child Well-being Explorer

2. DECLARATION ON THE USE OF AI TOOLS

AI tools were used strictly for coding assistance. Specifically, Claude Code was utilized to port the original R package ‘poseticDataAnalysis’ into Python to ensure structural alignment.

3. RESEARCH QUESTIONS

Is there a structural relationship between changes in public expenditure and shifts in child well-being

across European OECD countries? And if so, when analyzing these trade-offs through a non-compensatory framework, which countries emerge as leaders or laggards?

To answer these questions, this study moves beyond traditional single composite scores, demonstrating that a structural ordering approach captures the complexity of these relationships much more clearly.

3.1. Motivation and stakeholders

These questions are relevant for policy-makers, ministries, and international organisations that need to understand whether spending patterns are reflected in better child outcomes. They are also useful for civil society and monitoring bodies, because they help identify where public investment does, or does not, translate into measurable improvements in well-being.

The motivation for adopting a non-compensatory framework is both methodological and applied [1]. Composite indices, including those routinely produced for the OECD Better Life initiative

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and similar exercises, must collapse heterogeneous magnitudes onto a single line, which forces an implicit substitution rate between, for example, infant mortality and recreational expenditure per capita. These trade-offs rarely have ethical or policy meaning, and they obscure precisely the contradictions that policymakers most need to see: a country that excels in public spending while regressing on PISA outcomes is not “average” in any operationally meaningful sense, it is a country whose profile no scalar synthesis can faithfully represent [2–6]. A concrete instance of this opacity recurs in OECD Better Life dashboards, where small movements in a composite score can hide opposing movements on its constituent axes, and where two countries with the same composite value can occupy genuinely incomparable structural determinants of well-being [7].

Three categories of stakeholder benefit directly from a posetic reading of the same evidence. National educational ministries, particularly in the underperforming clusters that emerge from the cascade scores, require a way to identify which structural determinants of well being separate them from the leaders, indicator by indicator and macro-dimension by macro-dimension, rather than a black-box ranking. Treasury units in charge of family-spending budgets benefit from a separation matrix that quantifies how distant a country is from the Pareto frontier of public expenditure, since proximity to a maximal element is a more actionable target than a single score gap. International monitoring organisations and NGOs (for example UNICEF and the EU’s Better Internet for Kids policy network) need to detect partial regressions that would not surface in an aggregate index: this study shows that a country can advance on cognitive indicators while structurally regressing on the relational and digital ones, and that this dual movement is statistically visible only in the dominance space.

4. KEY-DEFINITIONS

- **Child Well-being:** A multidimensional phenomenon encompassing household economic conditions, physical health, educational opportunities, safety, social relationships, and subjective well-being.
- **Child Demographic:** The reference target for the measured indicators. While the pri-

mary focus rests on minors (0 - 17 years), the OECD framework adopts a life-cycle approach. The observed demographic spans from early infancy (e.g., infant mortality, vaccinations) through adolescence (e.g., 15-year-olds for PISA metrics) and extends into young adulthood (e.g., NEET rates measuring up to 29-year-olds) to capture the structural transition into independent socio-economic life.

- **Non-compensatory aggregation:** An evaluation in which a strong score on one dimension cannot offset a weak score on another, so that trade-offs (for example between health and public spending) are never silently averaged away [5, 6].
- **Incomparability:** The situation in which neither of two countries dominates the other, each being ahead on at least one indicator and behind on another; the framework keeps this status rather than forcing an artificial ranking.
- **Macro-areas:** A higher-level theme (material protection, health prevention, education and skills, social well-being) obtained by aggregating several related indicators while preserving their ordinal logic.

5. DATA DESCRIPTION

5.1. Data Source and Extraction

The dataset is extracted from the OECD Child Well-being (CWB) database. Relying on this authoritative institutional source ensures that the metrics utilized for comparative analysis are highly standardized, internationally comparable, and theoretically grounded [8].

5.2. Conceptual Framework and Indicator Selection

The original OECD framework organizes the indicators into three distinct conceptual macro-dimensions:

- *Child Outcomes (Group A):* Captures direct results affecting the child demographic, including physical health (e.g., infant mortality), educational performance (e.g., PISA top performers, NEET rates), and subjective life satisfaction.

- *Child Circumstances (Group B)*: Measures the contextual and environmental factors shaping child development, such as household relative income poverty, exposure to local crime, and sense of belonging at school.
- *Policies for Children (Group C)*: Quantifies structural and institutional inputs, including the length of statutory parental leave and general government expenditure on housing, health, and education.

From this framework, twenty-seven indicators were selected for the final analysis (see Appendix A).

5.3. Geographic and Temporal Scope

The study evaluates a targeted cross-section of sixteen European OECD countries: Sweden (SWE), Lithuania (LTU), Latvia (LVA), Estonia (EST), the United Kingdom (GBR), France (FRA), Luxembourg (LUX), Austria (AUT), Spain (ESP), Portugal (PRT), Italy (ITA), Slovenia (SVN), Poland (POL), the Czech Republic (CZE), Slovakia (SVK), and Hungary (HUN) (Figure 1).

The temporal scope focuses on the observation years 2015 and 2018.

The unit of analysis is the country by year pair, yielding thirty-two observations. To evaluate structural policy shifts and outcome trajectories over time, the 2015 and 2018 data are analysed separately; this separation allows the independent construction of year specific multidimensional representations, which are then aligned to compute temporal trajectories.

5.4. Strengths and Weaknesses

The dataset offers several strengths for comparative analysis. It rests on an authoritative institutional source (the OECD), which guarantees standardised and internationally comparable definitions. The framework is also multidimensional and theoretically grounded, capturing different facets of child well-being through both objective indicators (for example mortality rates and public expenditure) and subjective measures (for example PISA assessments and EU-SILC survey items). A further strength is the consistent country identification (ISO 3166 alpha-3) across all waves and indicators, which removes any reconciliation step before the analysis.

The dataset also carries clear limitations.

- *Temporal and geographic restrictions*: temporal coverage is narrow, since the 2009 and 2012 waves were excluded for severe missingness (about 42% of values in 2009 and 29% in 2012, against roughly 5% and 9% in 2015 and 2018), leaving only two cross sections separated by three calendar years. Geographic coverage is similarly reduced, with fourteen countries excluded for data-quality reasons, which limits representativeness.
- *Methodological and demographic heterogeneity*: the data combine distinct sources such as PISA, Eurostat and EU-SILC, and OECD Health Statistics, each with its own sampling design: for this reason years 2009, 2012, 2015, and 2018 were selected to align with the triennial data collection schedule of several key indicators; also the demographic reference varies through sources (PISA items concern 15 year old students, while transitional measures such as the NEET rate reach up to 29 years of age).
- *Broad policy metrics*: several policy inputs are general per-capita measures rather than expenditure targeted specifically at young people.
- *Residual missing values*: even after careful cleaning, some missing values remain, but these are handled within the poset framework through interval dominance rather than artificial point imputation.

6. METHODOLOGY

The analysis follows a stepwise procedure designed with a single foundational constraint: to preserve the inner structure of the data, indicators must be free of arbitrary weights and compensatory logics. To do this, we used poset as a base tool so that we could create a partial ranking among the nations using the indicators in the dataset [9, 10]. The twenty-seven indicators are cleaned and the ones with negative direction are oriented so that higher values always represent better conditions, then to match the non-compensatory research objective, they are partitioned into two separate analytical domains.

The first domain, child outcomes and circumstances, comprises twenty indicators (Groups A and B); the second, public expenditure and poli-

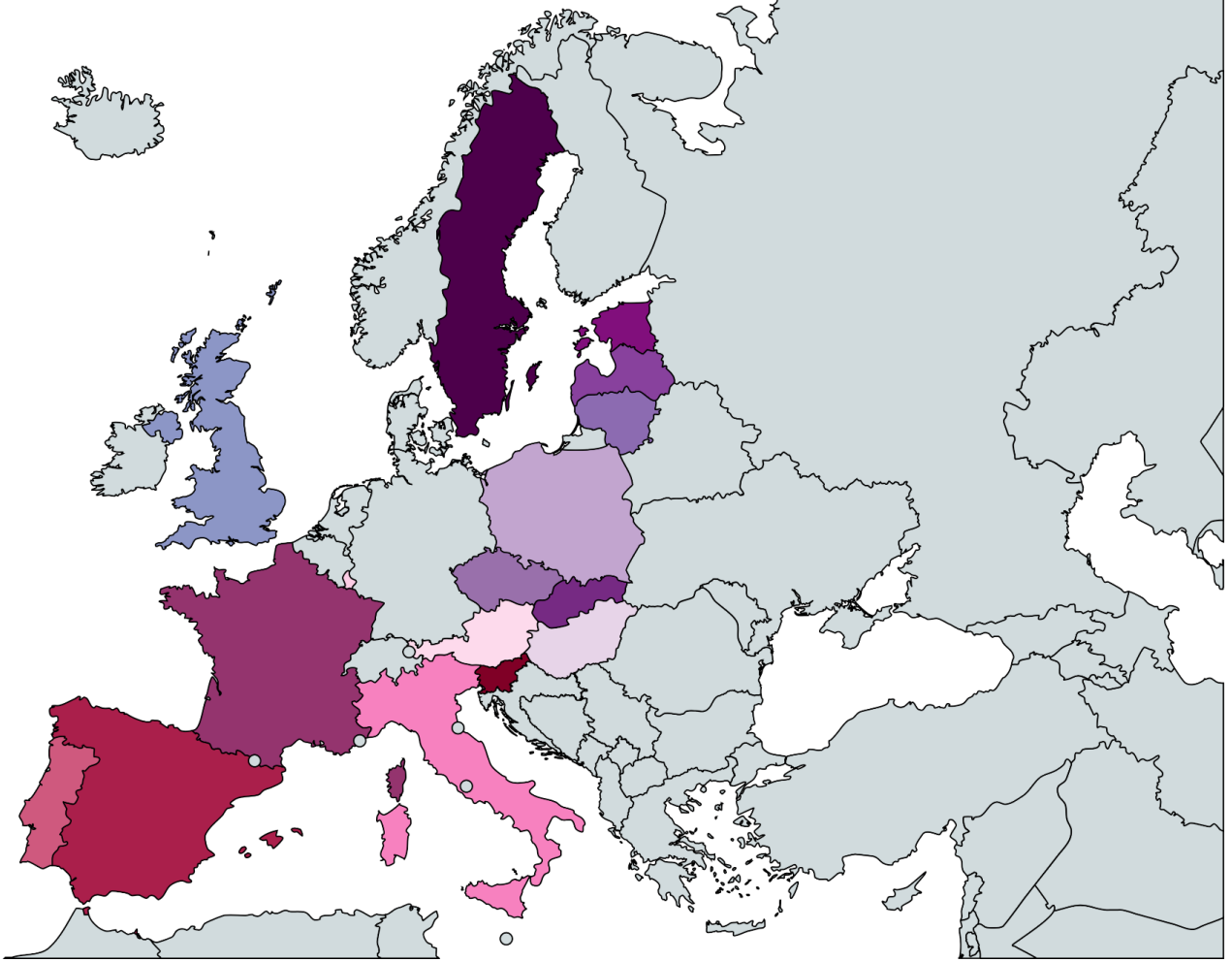


Figure 1: Map of the sixteen European countries selected for the youth well-being analysis. Map created using MapChart (<https://mapchart.net/>)

cies, comprises seven indicators (Group C). The two domains are analysed in parallel because they answer different questions, namely what young people experience versus what governments spend [1].

The first key step is dimension reduction performed by hierarchical aggregation: a single poset on the twenty outcome-and-circumstance indicators has demonstrated to dissolve into near-total incomparability, because the dimensionality is too high relative to sixteen countries for meaningful dominance to emerge [11, 12]. The MRP cascade is therefore introduced as a controlled, two-step aggregation that keeps poset logic at every stage while reducing the effective dimensionality, first to eleven macro-dimensions and then to four macro-areas (material protection, health prevention, education and skills, social well-being) for the outcome and circumstance block, and to four expenditure categories for the policy block (see

Appendix A).

During the aggregation phase, the missing values are treated as structural uncertainty and propagated through the process, rather than being imputed (see Appendix B).

To compare 2015 and 2018, the study uses Minimum Dominating Sets and poset separation. These tools show which countries remain on the frontier, which ones move away from it, and which ones change their relative position in the European structure. The separation information is then projected into a common two-dimensional map through Classical Multidimensional Scaling, and the two yearly maps are aligned with Procrustes rotation. This allows us to track country trajectories across time in a way that is easy to read and directly linked to the dominance structure.

In practical terms, short movements indicate stability, while long movements signal that a coun-

try's relative standing has changed substantially between the two waves.

7. ANALYTICAL TOOLS

7.1. Partial Ordered Set (poset) and Hasse diagrams

Used to show which countries are directly comparable and which ones are not, without forcing an artificial ranking. This is the first map of the European landscape: it identifies the countries that sit on the frontier, those that are clearly dominated, and those whose profiles are genuinely different and therefore not reducible to a single score.

7.2. Mutual Ranking Probability (MRP)

Used to transform the partial order into a readable ranking while keeping the non-compensatory logic of the analysis. MRP is the tool that makes the aggregated well-being and expenditure dimensions interpretable: it shows where each country stands within its peer group, but without introducing arbitrary weights or hiding trade-offs. It is also the basis for the progressive aggregation of the indicators into 11 and then 4 well-being dimensions, and from 7 to 4 expenditure dimensions.

7.3. Minimum Dominating Set (MDS)

Used to identify the smallest group of countries that are not dominated by others in the poset. This is the clearest summary of the frontier: it tells policy makers which countries represent the strongest structural profiles and how many different "models" of excellence exist in each year. Comparing the MDS across 2015 and 2018 also shows whether the frontier is stable or changing over time.

7.4. Posetic separation

Used to measure how far apart two countries are in the dominance structure, even when they are formally incomparable. This matters because two countries can both be non-dominated but still be very close or very far apart in practical terms; the separation matrix makes that difference visible. It is especially useful for understanding which countries are structurally similar and which ones are genuinely distant in their profile.

7.5. Classical MDS (cMDS) and Procrustes alignment

Used to project the separation matrix into a two-dimensional map and compare the 2015 and 2018 structures in the same space. This is the most intuitive visual output of the analysis: it shows clusters, outliers, and movements over time in a form that is easy to read and communicate. The Procrustes step ensures that movements across years reflect real structural change, not just differences in the way the maps are drawn.

7.6. Bottleneck analysis

Used to explain why two countries are not comparable in the dominance order. This is the diagnostic tool: instead of just saying that a country is behind, it shows which dimensions are blocking dominance and therefore where intervention would matter most.

The repository with the code used for this study is available on github [13].

8. RESULTS

For country profiles resulting from the poset analysis, please refer to Appendix C.

8.1. Well-being structure in 2015

In 2015, the well-being poset shows a clear but not fully linear hierarchy. The set of top-performing countries is formed by Estonia and Sweden, two countries that are incomparable because each leads on a different subset of dimensions: Estonia is stronger in education and social well-being, while Sweden leads in health prevention; both are equally strong in material protection. This means that neither country can be said to dominate the other in a complete sense, which is important for interpretation because it shows two distinct but equally strong models of child well-being.

Below the frontier, the structure remains informative and policy-relevant. Estonia is the most dominant country in the sample, followed by Sweden and Austria, while Italy and Poland occupy the lowest position with identical profiles. This bottom cluster is especially useful for stakeholders because it identifies countries that share the same structural weakness across all four well-being dimensions, rather than just having a low average score. (Figure 2a)

8.2. Well-being structure in 2018

By 2018, the frontier becomes broader and more differentiated. Sweden remains in the group of leading countries, joined by Portugal and Austria, and the three countries are now mutually incomparable because each combines strengths and weaknesses in a different way. Sweden is strongest on material protection, health, and education, Portugal leads on education and social well-being, and Austria is strongest on material protection and social well-being but weaker in health. It shows that the European well-being frontier did not simply shift upward or downward; it reconfigured. The frontier gained one additional country compared with 2015, and the overlap between the two years is limited to Sweden only, which indicates a partial turnover in leadership rather than a stable elite. (Figure 2b)

8.3. Changes over time

The year-to-year movements confirm that the sample is not static. Estonia records the largest decline, dropping sharply out of the frontier as social well-being falls substantially and education weakens as well. Portugal shows the strongest improvement, moving from a lower position in 2015 to the frontier in 2018, driven by gains in education and social well-being. Sweden remains the most stable leader, while Austria enters the frontier despite a slight decline in its overall score, showing that structural position depends on the full profile rather than on a single aggregate trend.

These movements are important for stakeholders because they separate two different questions: whether a country improves in absolute terms, and whether it changes its relative position in the European hierarchy. The analysis shows that those two questions do not always lead to the same answer.

8.4. Public expenditure structure

The expenditure poset is more stable and more concentrated at the top. In both years, the highest-performing group is made up of France and Luxembourg, which share the same maximum profile across the four expenditure dimensions. This creates a very stable leadership structure, with no change in the frontier between 2015 and 2018. The second tier is also fairly stable, with Sweden and Austria close behind, while lower-spending

countries remain grouped near the bottom. At the same time, the internal structure of the expenditure poset shows that countries do not follow a single fiscal model. The Czech Republic shifts from a highly specialised profile to a more balanced one, while Hungary moves in the opposite direction by reinforcing housing spending and reducing other categories. (Figure 3)

8.5. Expenditure and outcomes

The relationship between expenditure and well-being is weak and inconsistent in this period. France and Luxembourg remain at the top of the expenditure ranking, but their well-being trajectories diverge: France declines, while Luxembourg improves. Portugal shows the opposite pattern: it stays at the lowest expenditure level but improves strongly in well-being and reaches the well-being frontier. Estonia also shows that high well-being can coexist with low expenditure, but that this position may not remain stable over time. The broader message is structural rather than prescriptive. In this sample and time window, the countries with the highest public expenditure are not the same as those with the best child well-being outcomes, and changes in expenditure do not map cleanly onto changes in well-being. This suggests that public spending alone is not sufficient to explain the observed hierarchy, even though it remains one of the relevant dimensions of analysis. (Figure 4)

9. COMMENTS AND INTERPRETATION

The overall picture that emerges from the analysis is one of structural diversity rather than convergence. Countries do not line up neatly on a single scale of child well-being or public expenditure; instead, they occupy different positions that reflect distinct policy and social profiles. This is exactly why a non-compensatory reading is useful: it shows that countries can be strong in some dimensions and weak in others without those differences being hidden by a single average score.

9.1. Well-being frontiers and leadership

The well-being frontier changes visibly between 2015 and 2018. In 2015, the frontier of leading countries consisted of Estonia and Sweden, while



Figure 2: Hasse diagrams of Well-being posets for year 2015 (left) and 2018 (right)

in 2018 it expanded to Sweden, Portugal, and Austria. The overlap between the two frontiers is limited: the Jaccard similarity is 0.25, which means that only one country, Sweden, remains in both frontier sets. This indicates that leadership in child well-being is not fixed, but partly reshuffled over a three-years window.

Examining other countries' profiles as examples: Estonia passed from $| 3, 2, 3, 3 |$ to $| 2, 2, 2, 1 |$ recording the sharpest decline in the sample, losing its previous dominant position; Portugal, on the other hand, passed from $| 1, 3, 2, 2 |$ to $| 2, 2, 3, 3 |$ showing the strongest improvement; Sweden remains stable at the top, further increasing its leading position by gaining 1 additional point in education while maintaining the other areas at the same level as the previous observation.

9.2. Expenditure patterns and outcomes

The expenditure poset is more stable than the well-being one. France and Luxembourg remain at the top in both years with a $| 3, 3, 3, 3 |$ profile, and the expenditure frontier does not change at all, with Jaccard similarity equal to 1.0. The number of certain dominance pairs is 86 in 2015 and 73 in 2018, which still indicates a structured hierarchy, but one that is more stable than the

well-being ranking. Below the frontier, Sweden and Austria occupy the second tier in both years, while countries such as the Czech Republic and Hungary show internal reshuffling across spending categories. The Czech Republic shifts from a specialized $| 3, 1, 1, 3 |$ profile to a more balanced $| 2, 2, 2, 2 |$ one, while Hungary moves in the opposite direction passing from a $| 3, 1, 2, 2 |$ profile to a $| 3, 1, 1, 1 |$ one, concentrating its expenses efforts in housing and culture, while reducing expenses on other domains. This shows that expenditure differences are not only about level but also about composition.

The most striking numerical contrast concerns the relationship between expenditure and well-being (see Figure 4). France and Luxembourg remain the top spenders throughout the period, yet their well-being trajectories diverge: France shows decreases both in material and social domains, while Luxembourg has gains both in material and education, despite slightly decreasing its health score. Portugal remains at the minimum expenditure profile, which is $| 1, 1, 1, 1 |$ in both years, yet its well-being showed the greatest improvement, as seen in the previous paragraph. Estonia also illustrates the disconnect: it stays very low on expenditure, and despite slightly increasing its housing and culture spendings, its well-being drops

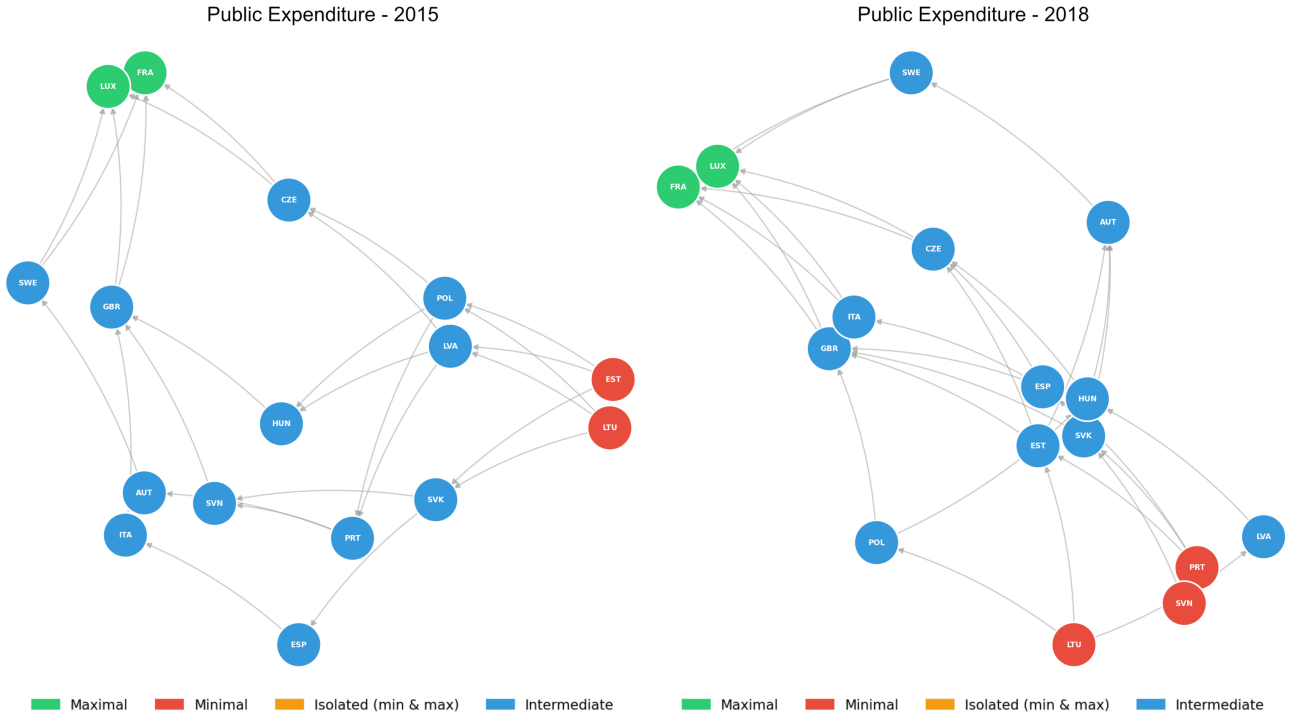


Figure 3: Hasse diagrams of Public Expenditure posets for year 2015 (left) and 2018 (right)

sharply from $|3, 2, 3, 3|$ to $|2, 2, 2, 1|$. These numbers suggest that spending alone does not determine the well-being hierarchy. In this sample, high expenditure is neither sufficient nor necessary for strong child well-being outcomes. What matters is the structure of the profile and how the different dimensions move together over time.

10. CONCLUSION

This work shows that European child well-being cannot be reduced to a single ranking without losing important structural information. The posetic approach makes it possible to distinguish countries that are genuinely leading from those that only appear similar under a composite score, and it shows that different countries can reach high performance through different combinations of strengths.

The well-being results reveal a clear change between 2015 and 2018. In 2015, the leading group was made up of Estonia and Sweden, while in 2018 the leading group expanded to Sweden, Portugal, and Austria. Sweden is the only country that remains at the top in both years, which makes it the most stable reference point in the sample. Estonia, by contrast, records the largest structural decline, driven above all by a strong fall in social well-being, while Portugal shows the strongest im-

provement and enters the leading group despite remaining at the lowest expenditure level.

The expenditure results tell a different story. France and Luxembourg remain the leaders in both years, with identical profiles and complete stability over time. This means that the expenditure hierarchy is much more fixed than the well-being hierarchy. At the same time, the expenditure structure does not match the well-being structure: the countries that spend the most are not the same as those with the best child well-being outcomes, and countries with similar spending levels can follow very different well-being trajectories.

This mismatch between expenditure and outcomes is the main policy-relevant finding. It suggests that public spending, at least as measured here, is not sufficient by itself to explain differences in child well-being across countries. Other factors outside the expenditure block must play an important role, including how policies are implemented and how different social and institutional contexts shape outcomes.

Overall, the report provides a structural reading of child well-being that is more informative than a standard composite index. It highlights leadership, stability, decline, and internal heterogeneity in a way that is directly useful for comparing coun-

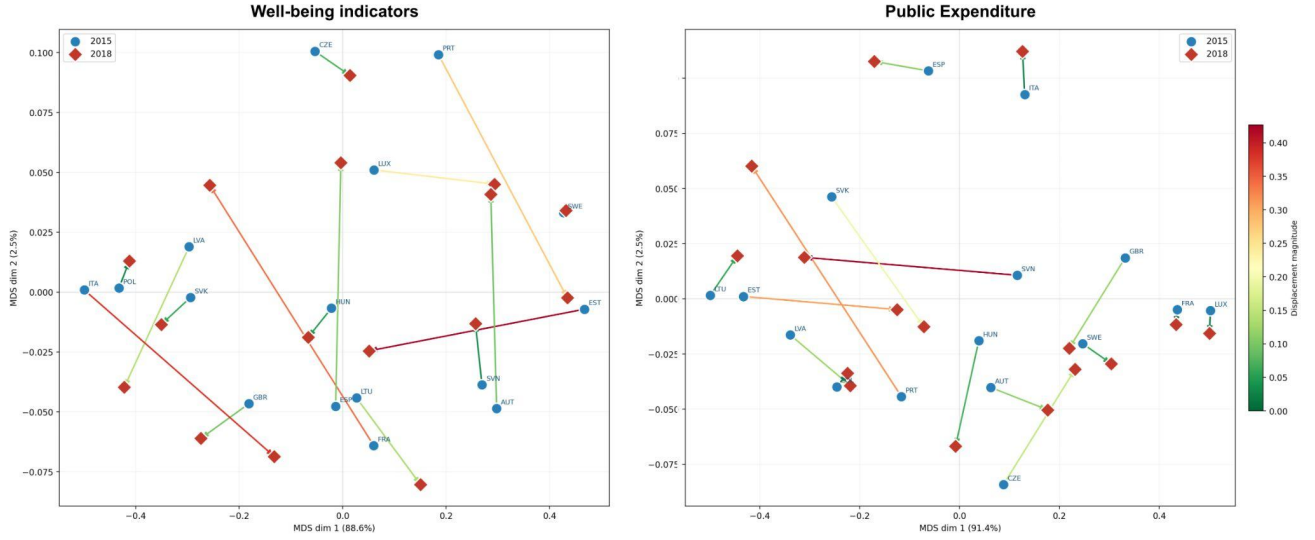


Figure 4: Country trajectories through the years 2015 and 2018 for Well-being indicators (left) and Public Expenditure (right)

tries and for understanding where the strongest differences lie.

The gap between the original ambition and the realised results is concentrated on the temporal axis. With only two retained waves separated by three calendar years, the longitudinal claims that can be made are bounded: the displacement vectors describe a snapshot of change, not a trend. The high missingness of the 2009 and 2012 waves made their inclusion methodologically unsafe, since the interval representation of nulls would have expanded the dominance regions of several countries to the point of dissolving the partial order. This is a data-quality constraint imposed by the OECD source rather than a design choice of the analysis, but it is the principal frontier for any extension of this work. A study covering more waves would be able to discriminate between transitory and persistent components of the structural shifts identified here.

The weaknesses of the present work, beyond the temporal sparsity already discussed, are mostly inherited from the OECD framework itself. The policy block (Group C) intermixes per-capita expenditure series and rate-type indicators, which the cascade treats uniformly once they are discretised but which are conceptually heterogeneous. Some PISA-derived indicators in the A and B dimensions reflect adolescent self-report and are sensitive to translation and cohort effects in ways that administrative records are not. The discretisation step in the cascade, while necessary for the partial order to admit meaningful structure on

sixteen units, introduces a cohort-relative element that interacts with the absolute movement of raw indicators (as the Estonia case study makes clear). None of these issues compromises the validity of the poset conclusions, but they limit the granularity at which causal narratives can be attached to the structural movements observed.

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Appendix

A. DATASET DESCRIPTION

Selected 27 indicators from the original OECD dataset used for the child well-being analysis project divided into well-being dataset and public expenditure dataset with the hierarchical aggregation (HA).

HA (1)	HA (2)	Code	Description	Unit	Direction
A1	MD1	A1_2	Children experiencing severe housing deprivation	%	negative
B1		B1_1	Children living in relative income poverty	%	negative
		B1_5	Students who firmly report that their parents encourage them to be confident	%	positive
C1		C1_2	Difference between before- and after-tax and transfer child relative income poverty rates	pp	positive
		C1_3	Guaranteed minimum income for a jobless couple with two children	%	positive
		C1_4	Total length of paid maternity and parental leave available to mothers	weeks	positive
		C1_5	Total length of paid paternity and parental leave reserved for fathers	weeks	positive
A2	MD2	A2_1	Infant mortality rates	Per 1 000 births	negative
C3		C3_2	Children <1 year old who received three doses of DPT vaccine	%	positive
		C3_3	Children <1 year old who received at least one dose of measles-containing vaccine	%	positive
A3	MD3	A3_3	Top performers in reading, maths and/or science	%	positive
		A3_4	Students who expect to complete tertiary education	%	positive
		A3_5	Youth not in education, employment or training (NEET)	%	negative
C4		C4_2	Typical net childcare costs for parents using centre-based care, two children in full-time care, two-earner couple on low earnings	% income	negative
		C4_6	Students to teaching staff in secondary education (public and private), based on full-time equivalents	ratio	negative
A4	MD4	A4_6	Students who report high satisfaction with their life as a whole	%	positive
B2		B2_4	Students who report experiencing bullying at school	%	negative
		B2_5	Students who feel like they belong at school	%	positive
B3		B3_5	Children in households that report crime and violence in their local area	%	negative
B4		B4_3	Students who firmly believe the Internet is a great resource for information	%	positive

HA	Code	Description	Unit	Direction
C2	C2_1	General government expenditure on housing and community amenities per person	USD PPP	positive
	C2_2	General government expenditure on recreation, culture and religion per person	USD PPP	positive
	C2_3	General government expenditure on housing social protection per person	USD PPP	positive
C3	C3_1	Government and compulsory contributory health insurance expenditure on health per person	USD PPP	positive
C4	C4_4	Public expenditure on primary, secondary and post-secondary non-tertiary education per full-time equivalent student	USD PPP	positive
	C4_5	Public expenditure on ancillary education services in primary, secondary and post-secondary non-tertiary education per full-time equivalent student	USD PPP	positive
C5	C5_1	General government expenditure on environment protection per person	USD PPP	positive

B. MISSING VALUES

After the initial filtering steps, the remaining missing values were not treated as entries to be imputed, but as a source of structural uncertainty. In practice, a unit with one or more null ordinal values does not correspond to a single point in the dominance space, but to the set of all profiles compatible with the observed information. For each country a , the observed profile is therefore represented as an interval in the hyperlattice, $[a_{\text{lower}}, a_{\text{upper}}]$, where a_{lower} replaces each null with the minimum admissible value and a_{upper} replaces each null with the maximum admissible value. Dominance is then evaluated in interval form: a is certainly below b if $a_{\text{upper}} \leq b_{\text{lower}}$, and possibly below b if $a_{\text{lower}} \leq b_{\text{upper}}$; otherwise, the two units remain incomparable. This approach allows all residual nulls to enter the aggregation as ordinal uncertainty rather than as arbitrary point values. Since the missingness is limited to a small number of indicators within multi-indicator subdimensions, the resulting uncertainty is contained, and the MRP scores obtained on this interval-based structure are typically very close to those that would arise from a median imputation, but without introducing an external assumption about the missing values.

C. COUNTRY PROFILES

The four well-being macro-dimension country profiles for 2015 are shown in the table below (levels 1–3, higher = better well-being)

Country	Material	Health	Education	Social
EST	3	2	3	3
SWE	3	3	2	2
AUT	3	1	3	3
LTU	2	1	3	3
SVN	2	2	2	3
ESP	2	2	1	3
LUX	2	3	1	2
PRT	1	3	2	2
CZE	1	3	2	1
FRA	3	1	1	2
HUN	2	2	2	1
GBR	3	1	1	1
LVA	1	1	2	1
SVK	1	1	1	2
ITA	1	1	1	1
POL	1	1	1	1

The four well-being macro-dimension country profiles for 2018 are shown in the table below (levels 1–3, higher = better well-being)

Country	Material	Health	Education	Social
SWE	3	3	3	2
PRT	2	2	3	3
LUX	3	2	2	2
AUT	3	1	2	3
SVN	2	2	2	3
LTU	1	1	3	3
CZE	3	2	1	1
ESP	3	1	1	2
EST	2	2	2	1
HUN	1	2	1	2
GBR	1	1	2	1
ITA	1	1	2	1
LVA	1	1	2	1
FRA	2	1	1	1
POL	2	1	1	1
SVK	1	1	1	2

The country profiles for public expenditures in 2015 are shown in the table below (levels 1–3, higher = better well-being)

Country	Housing/Culture	Public Health	Education	Environment
FRA	3	3	3	3
LUX	3	3	3	3
SWE	3	3	3	1
GBR	2	2	3	3
AUT	2	3	3	1
CZE	3	1	1	3
HUN	3	1	2	2
ITA	1	2	2	3
SVN	2	1	2	2
ESP	1	2	2	2
SVK	1	1	2	2
LVA	2	1	1	1
POL	2	1	1	1
EST	1	1	1	1
LTU	1	1	1	1
PRT	1	1	1	1

The country profiles for public expenditures in 2018 are shown in the table below (levels 1–3, higher = better well-being)

Country	Housing/Culture	Public Health	Education	Environment
FRA	3	3	3	3
LUX	3	3	3	3
SWE	3	3	3	1
AUT	3	3	3	1
GBR	2	2	3	2
CZE	2	2	2	2
ITA	1	2	2	3
ESP	1	2	1	2
HUN	3	1	1	1
POL	2	1	2	1
LVA	2	1	1	1
EST	2	1	1	1
SVN	1	1	2	1
SVK	1	1	2	1
LTU	1	1	1	1
PRT	1	1	1	1